

Computer Networks (6CS105CC26) — Sessional Exam Analysis

MCA Semester I, Nirma University | Past-paper and Question-Bank analysis restricted to your sessional syllabus

Scope and sources

Item	Detail
Sessional scope (from your 231-slide syllabus deck)	Unit 1 — Introduction to Data Communication & Networks: components, data flow, topologies, PAN/LAN/MAN/WAN, wireless, switching overview, Internet (slides 1–58) Unit 2 — Data Link Layer: framing, flow & error control, Stop-and-Wait, Go-Back-N (Ch. 11, slides 59–116) Unit 3 — Data & Signals: analog/digital, bandwidth, bit rate, transmission impairment (Ch. 3, slides 117–196) Unit 4 — Switching: circuit, datagram, virtual-circuit networks (Ch. 8, slides 197–231)
Papers analysed	Semester-End Jan 2022 (50 marks, 2 h), Jan 2023, Jan 2025 and Dec 2025 (100 marks, 3 h each), plus the 58-item Question Bank (6CS105CC26).
How to read “frequency”	Number of the 4 past papers in which the topic appeared. The Question Bank (QB) column is a second, independent signal. Unit names are my labels based on chapters; your deck does not number units.

Key findings at a glance

- **Past finals under-represent your sessional.** Only 6–22% of the marks in each past paper sit clearly inside your slides (Table 1). Finals are weighted towards OSI, IP addressing, TCP, MAC protocols and Hamming code, which are not in your deck.
- **The Question Bank is your best guide:** 28 of its 58 questions fall inside your sessional slides (Q1–9, Q19–23, Q30–39, Q55–58).
- **ARQ is the most repeated topic:** Selective Repeat vs Go-Back-N appears in 3 of 4 papers, in near-identical wording twice (Jan 2023, Jan 2025).
- **Framing (bit/byte stuffing) and topologies** each appear in 2 of 4 papers. Signal numericals are low-effort marks and follow a reusable template.
- **Gap to fix:** your deck mentions Selective Repeat ARQ on only 3 slides, so study it from the textbook (the slides appear to follow Forouzan chapter numbering). Switching (Ch. 8) never appeared in the 4 finals but has 4 QB questions.

Table 1 — How much of each past paper falls in your sessional scope

Past paper	Max marks	Clearly in sessional slides	Partly covered (adjacent / briefly in slides)	Clear share	Clear + partial share
Semester-End, Jan 2022	50	11 marks — Q1 (8) PAN/LAN/MAN/WAN; Q8 (3) bandwidth	8 marks — Q4 OR option: Selective Repeat ARQ	22%	38%
Semester-End, Jan 2023	100	10 marks — Q3A (4) mesh calculation; Q3B (6) bit stuffing	14 marks — Q1B (8) SR vs GBN; Q2A (6) unicast/multicast/broadcast	10%	24%
Semester-End, Jan 2025	100	8 marks — Q1b (8) Star/Bus/Mesh/Ring	6 marks — Q2c (6) SR vs GBN	8%	14%
Semester-End, Dec 2025	100	6 marks — Q2b (6) framing	≈4 marks (estimate) — data-link half of Q1a (8)	6%	10%

“Partly covered” = the topic is only briefly in your slides (Selective Repeat: slides 83, 105, 107; unicast/multicast not explained) or overlaps with your slides (data-link layer functions). Other questions in these papers (OSI layers, MAC protocols, Hamming, IP, TCP, DNS, etc.) fall outside your deck.

1. Topic Weightage Table (Unit · Topic · Frequency · Typical marks)

Unit	Specific topic	Slides	Where it appeared in past papers (marks)	Freq. (of 4)	Question Bank items	Typical marks	Priority
UNIT 1 — Introduction to Data Communication & Networks (slides 1–58)							
Unit 1	Network topologies: Mesh, Star, Bus, Ring (Hybrid and Tree also in slides)	23–38	Jan 2023 Q3A — cables & ports for a 5-device mesh (4) Jan 2025 Q1b — differentiate Star, Bus, Mesh, Ring with diagrams (8)	2 / 4	Q2, Q3, Q4, Q5 (5–8 each); Q1e	4–8	HIGH
Unit 1	Network categories: PAN, LAN, MAN, WAN (+ wireless networks)	39–52	Jan 2022 Q1 — key characteristics of PAN, LAN, MAN, WAN (8)	1 / 4	Q6 (2–3); Q7, Q8, Q9 (5–8 each)	3–8	HIGH
Unit 1	Core definitions: network, data communication, point-to-point vs multipoint, unicast / multicast / broadcast, analog vs digital, bandwidth	5–7, 21, 119–122	Jan 2023 Q2A — distinguish unicast, multicast, broadcast (6)	1 / 4	Q1(a–m): 13 short definitions	1–6	MED-HIGH

Unit	Specific topic	Slides	Where it appeared in past papers (marks)	Freq. (of 4)	Question Bank items	Typical marks	Priority
Unit 1	DC vs computer network, 5 components of DC, data flow (simplex / half / full duplex), protocol elements (syntax, semantics, timing), network criteria	6–20	Not asked directly	0 / 4	Q1(a, b) only	1–3	LOW
Unit 1	Switching overview, Internet, ways of accessing the Internet	53–57	Not asked	0 / 4	None	–	LOW
UNIT 2 — Data Link Layer: Framing, Flow & Error Control (Ch. 11, slides 59–116)							
Unit 2	Framing: byte stuffing, bit stuffing, fixed vs variable-size framing	60–66	Jan 2023 Q3B — illustrate bit stuffing with example (6) Dec 2025 Q2b — why framing is required + any 3 frame header fields (6)	2 / 4	Q30 (justify framing; bit vs byte), Q31, Q32	6	HIGH
Unit 2	Selective Repeat ARQ and SR vs Go-Back-N comparison	83, 105, 107 (SR only briefly mentioned)	Jan 2022 Q4-OR — how SR gives error & flow control, with example (8) Jan 2023 Q1B — why SR beats GBN; diagram + critical comments (8) Jan 2025 Q2c — same question, same wording (6)	3 / 4	Q35, Q37, Q38	6–8	HIGH
Unit 2	Go-Back-N ARQ and Stop-and-Wait ARQ: working, sequence numbers, timers, lost frame / lost ACK	78–116	Appears as the baseline inside the GBN-vs-SR questions (Jan 2023 Q1B, Jan 2025 Q2c)	2 / 4 (indirect)	Q33, Q34, Q36, Q37	6–8	HIGH
Unit 2	Sliding-window numericals: sender / receiver window size for a given maximum sequence number	107–116	Not asked	0 / 4	Q38, Q39	1–3 (est.)	MEDIUM
Unit 2	Flow control vs error control, data link control, noiseless-channel protocols (Simplest, Stop-and-Wait)	67–82	Dec 2025 Q1a — functions of Data Link layer (+ Application layer) (8; data-link half overlaps)	1 / 4 (partial)	Background for Q33	3–4	MEDIUM
UNIT 3 — Data & Signals (Ch. 3, slides 117–196)							
Unit 3	Bandwidth & frequency-domain numericals: periodic / non-periodic composite signals, draw spectrum	146–165	Jan 2022 Q8 — bandwidth 100 kHz, middle frequency 150 kHz, peak 10 V; draw frequency domain (3)	1 / 4	Q19, Q20 (2–4 each)	2–4	HIGH
Unit 3	Bit rate and signal-level numericals (bits per level, required bit rate)	166–175	Not asked	0 / 4	Q21 (1–2), Q22 (1–3)	1–3	MEDIUM

Unit	Specific topic	Slides	Where it appeared in past papers (marks)	Freq. (of 4)	Question Bank items	Typical marks	Priority
Unit 3	Transmission impairment: attenuation, distortion, noise; dB and SNR	180–196	Not asked	0 / 4	Q23 (3–6)	3–6	MEDIUM
Unit 3	Analog vs digital signals, sine-wave parameters (amplitude, period, frequency, phase), time vs frequency domain	119–145	Not asked	0 / 4	Q1(i–l)	1–3	LOW
UNIT 4 — Switching (Ch. 8, slides 197–231)							
Unit 4	Circuit switching: three phases, efficiency, delay; advantages & disadvantages	204–213	Not asked	0 / 4	Q55, Q56 (marks not stated)	5–8 (est.)	MEDIUM
Unit 4	Datagram (packet) switching, virtual-circuit switching, message switching	203, 214–231	Not asked	0 / 4	Q55, Q57, Q58 (marks not stated)	5–8 (est.)	MEDIUM

Priority combines past-paper frequency, Question Bank coverage and marks. “est.” = my estimate, because the Question Bank states no marks for those items. Zero appearances in a final paper does not mean unimportant for a sessional, since those papers cover the whole course. Unicast and multicast are not explained in your slides (only “broadcast” appears), so use the textbook for those.

2. High-Yield Priority List — study these 5 first

Rank	Topic	Evidence	What to prepare (exam-ready checklist)	Expected marks
1	ARQ protocols: Selective Repeat vs Go-Back-N (with Stop-and-Wait as base)	Only topic seen in 3 of 4 papers. Jan 2023 Q1B and Jan 2025 Q2c: near-identical wording (why SR performs better than GBN; justify with diagram and critical comments). Jan 2022 Q4-OR: SR error and flow control with an example. QB has 7 ARQ items (Q33–39).	- Timeline diagrams for Stop-and-Wait, GBN and SR showing a lost frame and a lost ACK - Comparison points: frames resent (all outstanding vs only the lost one), receiver window (1 vs N), out-of-order frames (discarded vs buffered), timers / NAK, bandwidth efficiency, receiver complexity - Window rules: GBN sender $2^m - 1$, receiver 1; SR both 2^{m-1} - Finish with a verdict: SR wastes less bandwidth but needs more buffer and logic - SR is only mentioned on 3 slides, so study it from the textbook	6–8 (a full sub-question)
2	Framing: byte stuffing and bit stuffing	2 of 4 papers, 6 marks each (Jan 2023 Q3B, Dec 2025 Q2b). QB Q30–32 cover justification, bit stuffing and byte stuffing.	- Why framing is needed (frame boundaries, receiver synchronisation); fixed vs variable-size framing - Byte stuffing: flag + ESC byte, with a worked stuff / unstuff example - Bit stuffing: flag 01111110, insert a 0 after five consecutive 1s, with a worked example - Table: bit vs byte stuffing; plus three header fields with purpose (e.g. address, control, length/type)	6
3	Network topologies: Mesh, Star, Bus, Ring	2 of 4 papers (Jan 2023 Q3A numerical, 4 marks; Jan 2025 Q1b diagram-based, 8 marks). QB Q2–5: one full question per topology.	- Neat diagram of each topology 4–5 advantages and disadvantages each (slides 24–34) - One comparison table: cost, cabling, reliability, fault isolation, single point of failure, scalability - Mesh formulas: links = $n(n-1)/2$ (duplex), $n(n-1)$ (simplex); ports per device = $n-1$	4–8
4	Network categories: PAN, LAN, MAN, WAN	Jan 2022 Q1 — the opening 8-mark question. QB Q6–9 (PAN 2–3 marks, others 5–8).	- For each: definition, area covered, ownership, speed, typical technology, example - One comparison table covering all four - Wireless categories (Bluetooth, WLAN / IEEE 802.11, wireless WAN) and WAN vs Internet difference	3–8

Rank	Topic	Evidence	What to prepare (exam-ready checklist)	Expected marks
5	Signal numericals: bandwidth, spectrum, bits per level, bit rate	Jan 2022 Q8 (3 marks) is QB Q20 with different numbers. QB Q19–22 are short textbook-style numericals.	- Bandwidth = $f_{\text{high}} - f_{\text{low}}$; draw spectrum with labelled axes - Bits per level = $\log_2(L)$; bit rate = total bits $\div$ time (state page-size assumptions) - Re-solve each QB numerical with changed values, since examiners reuse the template	1–4 each (easy marks)

If time remains: (a) Switching — compare circuit, datagram and virtual-circuit (QB Q55–58); (b) transmission impairment — attenuation, distortion, noise (QB Q23, 3–6 marks).

3. Question Patterns

3.1 Recurring question formats

Pattern	Typical wording	Seen in	Marks · CO · BL	How to answer
Why is X better than Y? (justify)	Why does Selective Repeat ARQ give better performance than Go-Back-N? Justify with a diagram and critical comments.	Jan 2023 Q1B, Jan 2025 Q2c (near-verbatim); Jan 2022 Q4-OR (variant)	6–8 · CO2 · BL3–BL5	Draw both protocols side by side with the same loss scenario, tabulate differences, end with a reasoned verdict.
Differentiate / compare	Differentiate Star, Bus, Mesh and Ring with diagrams; distinguish unicast, multicast, broadcast; key characteristics of PAN, LAN, MAN, WAN.	Jan 2025 Q1b; Jan 2023 Q2A; Jan 2022 Q1	6–8 · CO1 · BL2–BL3	Comparison table with 4–6 attributes plus a labelled diagram for each item.
Illustrate with an example	Illustrate bit stuffing with an example; illustrate SR ARQ by taking a suitable example.	Jan 2023 Q3B; Jan 2022 Q4-OR	6–8 · CO2 · BL3–BL4	Two-line definition, then spend most of the space on a worked example or timeline.
Why is X required? + enumerate	Why is framing required at the data link layer? Give reasons and explain any three header fields.	Dec 2025 Q2b	6 · CO1 · BL2	Give 2–3 reasons, then exactly three fields with a one-line purpose each.
Numerical with assumed data	Cables and ports for a 5-device mesh; draw the frequency domain of a composite signal.	Jan 2023 Q3A; Jan 2022 Q8; QB Q19–22, Q38–39	3–4 · CO1–2 · BL3	Formula → substitution → answer → neat sketch. State assumptions (papers say “assume suitable data”).
Define briefly	Define network topology, broadcast, bandwidth, analog signal, etc.	QB Q1(a–m)	1–3 each	2–3 lines each plus a one-word example.
Explain in detail + advantages / disadvantages	Explain mesh / star / bus / ring topology; explain circuit / message / packet switching.	QB Q2–5, Q56–58	5–8	Principle, diagram, 3–4 advantages, 3–4 disadvantages.

Pattern	Typical wording	Seen in	Marks · CO · BL	How to answer
Same template, new numbers	Jan 2022 Q8 (100 kHz, 150 kHz, 10 V) is QB Q20 (200 kHz, 140 kHz, 20 V) with new values.	Jan 2022 Q8 vs QB Q20	–	Solve every QB numerical, then re-solve with changed values. Across the finals, Hamming (4/4 papers), IP block (3/4), fragmentation (2/4) and routing table (2/4) were also reused with new numbers.

3.2 Paper-structure observations

Observation	Detail
Where sessional topics sit	In all three 100-mark papers they fall in Section I (Q1–Q3); Section II (Q4–Q6) has none. In the 50-mark Jan 2022 paper they are Q1, Q4 (OR) and Q8.
Choice	The 100-mark papers are fully compulsory (no optional questions). Only the 2-hour Jan 2022 paper offered OR options (Q4, Q5).
Mark blocks	Sub-questions carry 3, 4, 6 or 8 marks. The opening Q1 is usually two 8-mark parts; Q2 is usually three 6-mark parts.
Difficulty level	Mostly CO1–CO2 and BL2–BL3 (understand / apply). “Justify” questions rise to BL4–BL5 (Jan 2022 Q4 is BL4; Jan 2023 Q1B is BL5).
Diagrams expected	Almost every in-scope question needs a diagram or worked example: topology sketches, ARQ timelines, stuffing example, signal spectrum.

4. Quick-Answer Key for the Numericals (verify against your own working)

Question	Working	Answer
Jan 2023 Q3A — 5-device mesh: cables and ports	Cables = $n(n-1)/2 = 5 \times 4/2$. Each device needs $n-1 = 4$ ports; total ports = 5×4 .	10 cables; 4 ports per device (20 in total)
Jan 2022 Q8 — bandwidth 100 kHz, middle 150 kHz, peak 10 V	$f_{\text{low}} = 150 - 50 = 100$ kHz; $f_{\text{high}} = 150 + 50 = 200$ kHz. Non-periodic: continuous spectrum, amplitude 0 at both edges.	Bandwidth 100 kHz; curve from 100 to 200 kHz peaking at 10 V at 150 kHz
QB Q19 — five sine waves at 100, 300, 500, 700, 900 Hz	Bandwidth = $f_{\text{high}} - f_{\text{low}} = 900 - 100$. Draw five vertical lines of height 10 V.	800 Hz
QB Q20 — bandwidth 200 kHz, middle 140 kHz, peak 20 V	$f_{\text{low}} = 140 - 100 = 40$ kHz; $f_{\text{high}} = 140 + 100 = 240$ kHz; amplitude 0 at the two extremes.	Spectrum 40–240 kHz, peak 20 V at 140 kHz
QB Q21 — digital signal with 8 levels	Bits per level = $\log_2(8)$.	3 bits per level
QB Q22 — 100 pages per minute	Assume 24 lines $\times$ 80 characters $\times$ 8 bits = 15,360 bits per page; $\times 100$ pages = 1,536,000 bits per minute; $\div 60$ s.	25,600 bps = 25.6 kbps (state the assumption)
QB Q38 — Selective Repeat, maximum sequence number 15	Sequence numbers 0–15 means 16 numbers, so $m = 4$. Window = $2^{(m-1)}$.	Sender = receiver = 8

Question	Working	Answer
QB Q39 — Go-Back-N, maximum sequence number 15	$m = 4$. Sender window = $2^m - 1$; receiver window = 1.	Sender 15, receiver 1

Appendix A — Past-paper questions mapped to your sessional scope

Paper	Q	Question (condensed)	Marks	Topic	Scope status
Jan 2022	Q1	Key characteristics of PAN, LAN, MAN and WAN	8	Network categories	In scope
Jan 2022	Q4 (OR)	How Selective Repeat ARQ provides error and flow control; example	8	SR ARQ	Partial (SR briefly in slides)
Jan 2022	Q8	Composite signal: bandwidth 100 kHz, middle frequency 150 kHz, peak 10 V; draw frequency domain	3	Bandwidth numerical	In scope
Jan 2023	Q1B	Why SR ARQ performs better than Go-Back-N; diagram and critical comments	8	SR vs GBN	Partial (GBN in slides, SR briefly)
Jan 2023	Q2A	Distinguish unicast, multicast and broadcast	6	Core terminology	Partial (only broadcast in slides)
Jan 2023	Q3A	Cables and ports needed for 5 devices in mesh topology	4	Topology numerical	In scope
Jan 2023	Q3B	Illustrate bit stuffing with an example	6	Framing	In scope
Jan 2025	Q1b	Differentiate Star, Bus, Mesh, Ring topologies with diagrams	8	Topologies	In scope
Jan 2025	Q2c	Why SR ARQ performs better than Go-Back-N; diagram and critical comments	6	SR vs GBN	Partial (GBN in slides, SR briefly)
Dec 2025	Q1a	Key functions of Application layer and Data Link layer (OSI)	8	Layer functions	Partial (data-link half only)
Dec 2025	Q2b	Why framing is required at the data link layer; any three header fields	6	Framing	In scope

All other past-paper questions (OSI layers, MAC protocols, Hamming code, IP addressing and fragmentation, routing tables, TCP, DNS, VPN, ICMP, ARP/RARP, digital signature, fibre optic cable, bridge vs router) fall outside your sessional slides and are excluded from the analysis.

Appendix B — Question Bank items with no matching slides in your deck

QB items	Topics	Status
Q10–Q18	Layers / protocols / interfaces, design issues, connection-oriented vs connectionless, reliable vs unreliable service, OSI and TCP/IP models, standardisation	No slides cover these. OSI appears in 3 of 4 past papers, so if your faculty included it in the sessional, add it to your plan.
Q24–Q29	Internet checksum, CRC, Hamming code	Error detection and correction is not in the deck (Hamming appears in all 4 papers). Confirm with faculty.

QB items	Topics	Status
Q40–Q54	ALOHA, CSMA / CD / CA, controlled access, channelization (FDMA, TDMA, CDMA)	Not in the deck (medium access control). FDM / TDM are only mentioned in the circuit-switching slide.